

Logistic Regression

COMP 5212: Machine Learning

From linear prediction to class probabilities

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A probability model determines what a good fit means.

Learning Goals

By the end of the lecture, you should be able to:

- formulate logistic regression for binary outcomes;
- derive least squares and logistic loss from likelihood;
- derive and apply the gradient-descent update;
- explain softmax regression and its categorical likelihood;
- recognize softmax in classifiers, language models, and attention.

Same inputs, a different prediction

Linear regression

Sensor readings x

→ electricity consumption $Y \in \mathbb{R}$

$$\hat{y} = b + w^T x.$$

Binary classification

Sensor readings x

→ defective or acceptable $Y \in \{0, 1\}$

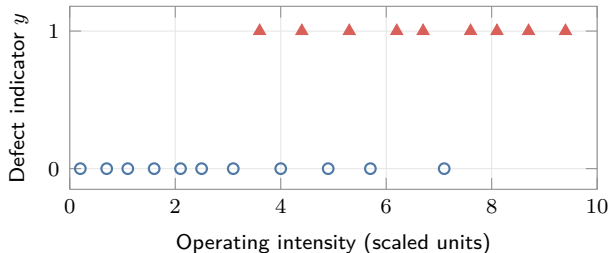
$$\pi(x) = \Pr(Y = 1 \mid x).$$

What should replace a real-valued prediction?

Predict a probability first; use it to make a decision.

Motivation: detecting defective parts

A sensor measures process intensity. Inspection records whether each part is defective



x : sensor measurement

$y = 1$: defective

$y = 0$: acceptable

What should we predict for the next part?

$$\Pr(Y = 1 \mid x).$$

A probability expresses uncertainty.

Probabilities and decisions

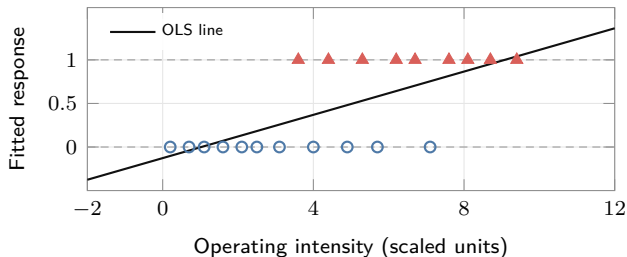
Part	Predicted defect probability	Decision at threshold 0.5
A	0.55	Flag
B	0.95	Flag

Do these predictions carry the same information?

- Probabilities help rank inspection priorities.
- A decision also depends on the cost of each mistake.

First estimate a probability. Then choose the action appropriate to the application.

OLS with binary responses



For $Y \in \{0, 1\}$,

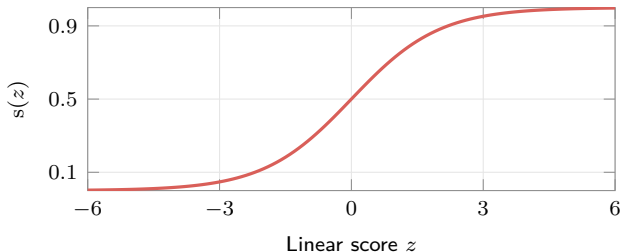
$$\mathbb{E}[Y \mid x] = \Pr(Y = 1 \mid x).$$

An unconstrained line can predict below 0 or above 1.

How can a linear score become a valid probability?

OLS can fit binary labels. Its unrestricted linear predictions need not respect probability bounds.

The logistic function



$$s(z) = \frac{1}{1 + e^{-z}}.$$

- $s(0) = 1/2$.
- $0 < s(z) < 1$ for finite z .
- $s(-z) = 1 - s(z)$.

$$s'(z) = s(z)(1 - s(z)).$$

The score is unbounded. The probability stays between zero and one.

A linear model for log-odds

For $0 < \pi(x) < 1$, odds are $\pi(x)/(1 - \pi(x))$

$$\log \frac{\pi(x)}{1 - \pi(x)} = b + w^T x = z,$$

$$\frac{\pi(x)}{1 - \pi(x)} = e^z,$$

$$\pi(x) = \frac{e^z}{1 + e^z} = \boxed{s(b + w^T x)}.$$

Logistic regression assumes that log-odds are linear in the chosen features.

Why start with a linear score?

Suppose the true probability $q(x)$ is smooth and strictly between 0 and 1

$$g(x) = \log \frac{q(x)}{1 - q(x)}.$$

$$g(x) \approx g(x_0) + \nabla g(x_0)^T (x - x_0) = b + w^T x.$$

A local approximation to the log-odds gives a logistic model for the probability.

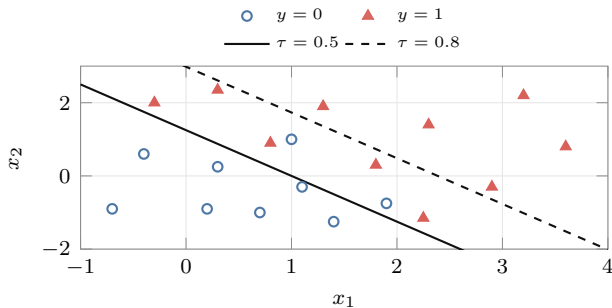
A useful starting model; its adequacy still needs to be checked on data.

Model and notation

$$\tilde{x}_i = \begin{bmatrix} 1 \\ x_i \end{bmatrix}, \quad \theta = \begin{bmatrix} b \\ w \end{bmatrix}, \quad X = \begin{bmatrix} \tilde{x}_1^\top \\ \vdots \\ \tilde{x}_n^\top \end{bmatrix} \in \mathbb{R}^{n \times p},$$
$$z = X\theta, \quad \pi_i = s(z_i), \quad \pi = (\pi_1, \dots, \pi_n)^\top.$$

- p counts all fitted coefficients, including the intercept.
- $y_i \in \{0, 1\}$ and $\pi_i = \Pr(Y_i = 1 \mid \tilde{x}_i; \theta)$.
- The feature vector can contain transformed measurements.

Decision thresholds and boundaries



For $0 < \tau < 1$, predict class 1 if $\pi(x) \geq \tau$.

$$b + w^T x \geq \log \frac{\tau}{1 - \tau}.$$

$\tau = 0.5$: score boundary 0.

$\tau = 0.8$: score boundary $\log 4$.

Changing the threshold moves the decision boundary without refitting the probability model.

Worked example: score to decision

$z = -2 + 0.8x$, $\pi = s(z)$, decision threshold $\tau = 0.7$

x	Score z	Probability π	Predicted class
2	-0.4	0.401	0
4	1.2	0.769	1

At what sensor value does the decision change?

$$x = \frac{2 + \log(0.7/0.3)}{0.8} \approx 3.559.$$

What do we learn from the data?

$$\mathcal{D} = \{(x_i, y_i)\}_{i=1}^n, \quad y_i \in \{0, 1\}, \quad \pi_i(\theta) = s(\tilde{x}_i^\top \theta).$$
$$\hat{\theta} = \arg \min_{\theta \in \mathbb{R}^p} J(\theta).$$

Which objective rewards good probability predictions?

Compare the probability assigned to each observed label.

Maximum likelihood turns this requirement into a loss function.

What does conditional likelihood measure?

We observe the inputs X and responses y . A model assigns them

$$\mathcal{L}(\theta; y \mid X) = f_{\theta}(y \mid X).$$

As a likelihood, this is a function of θ , with X and y held fixed.

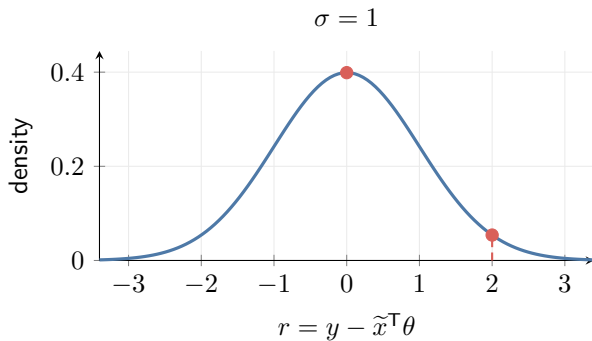
If the responses are independent conditional on X ,

$$f_{\theta}(y \mid X) = \prod_{i=1}^n f_{\theta}(y_i \mid \tilde{x}_i).$$

Maximum likelihood chooses parameters that give the observed responses the largest joint probability mass or density, within the chosen model.

Gaussian errors assign density to residuals

For this comparison, first let the response be real-valued: $y_i \in \mathbb{R}$.



$$Y_i = \tilde{x}_i^T \theta + \varepsilon_i,$$

$$\varepsilon_i \mid X \stackrel{\text{iid}}{\sim} \mathcal{N}(0, \sigma^2), \quad \sigma > 0.$$

For the observed response, the residual is

$$r_i(\theta) = y_i - \tilde{x}_i^T \theta.$$

At fixed σ , a larger $|r_i|$ receives a smaller Gaussian density.

The Gaussian likelihood leads to squared error

$$\mathcal{L}(\theta, \sigma; y \mid X) = \prod_{i=1}^n \frac{1}{\sqrt{2\pi\sigma^2}} \exp\left[-\frac{(y_i - \tilde{x}_i^\top \theta)^2}{2\sigma^2}\right].$$
$$-\log \mathcal{L} = \underbrace{\frac{n}{2} \log(2\pi\sigma^2)}_{\text{constant in } \theta} + \frac{1}{2\sigma^2} \underbrace{\sum_{i=1}^n (y_i - \tilde{x}_i^\top \theta)^2}_{\text{SSE}(\theta)}.$$

For any fixed $\sigma > 0$, maximizing this likelihood over θ is equivalent to

$$\min_{\theta} \text{SSE}(\theta) = \min_{\theta} \|y - X\theta\|_2^2.$$

Differentiating gives the normal equations

Use the convenient scaling $J_{\text{LS}}(\theta) = \|y - X\theta\|_2^2/(2n)$:

$$J_{\text{LS}}(\theta) = \frac{1}{2n} (y^\top y - 2\theta^\top X^\top y + \theta^\top X^\top X \theta).$$

$$\nabla J_{\text{LS}}(\theta) = \frac{1}{n} (X^\top X \theta - X^\top y) = \frac{1}{n} X^\top (X \theta - y).$$

$$\nabla J_{\text{LS}}(\hat{\theta}) = 0 \quad \Longleftrightarrow \quad \boxed{X^\top X \hat{\theta} = X^\top y}.$$

The objective is convex, so every solution of these equations minimizes it.

Are the least-squares coefficients unique?

If $\text{rank}(X) = p$, then $X^T X$ is invertible and the solution is unique:

$$\hat{\theta} = (X^T X)^{-1} X^T y.$$

If $\text{rank}(X) < p$, every least-squares solution has the form

$$\hat{\theta} = X^\dagger y + v, \quad v \in \ker(X).$$

Here $X^\dagger$ is the Moore–Penrose pseudoinverse.

$X^\dagger y$ is the unique minimum-norm coefficient vector. All least-squares solutions give the same fitted responses $XX^\dagger y$.

The probability of an observed binary label

Now $y_i \in \{0, 1\}$ and

$$Y_i \mid \tilde{x}_i; \theta \sim \text{Bernoulli}(\pi_i), \quad \pi_i = s(\tilde{x}_i^\top \theta).$$

The model assigns the observed label probability

$$\Pr(Y_i = y_i \mid \tilde{x}_i; \theta) = \begin{cases} \pi_i, & y_i = 1, \\ 1 - \pi_i, & y_i = 0. \end{cases}$$

$$\Pr(Y_i = y_i \mid \tilde{x}_i; \theta) = \boxed{\pi_i^{y_i} (1 - \pi_i)^{1-y_i}}.$$

Bernoulli likelihood and binary cross-entropy

Conditional independence gives

$$\mathcal{L}(\theta; y \mid X) = \prod_{i=1}^n \pi_i^{y_i} (1 - \pi_i)^{1-y_i}.$$

$$\log \mathcal{L}(\theta) = \sum_{i=1}^n [y_i \log \pi_i + (1 - y_i) \log(1 - \pi_i)].$$

$$J(\theta) = -\frac{1}{n} \log \mathcal{L}(\theta) = \frac{1}{n} \sum_{i=1}^n \underbrace{[-y_i \log \pi_i - (1 - y_i) \log(1 - \pi_i)]}_{\text{binary cross-entropy } \ell_i(\theta)}.$$

Natural logarithms are used throughout. Averaging preserves any unpenalized optimum.

Express the same loss through the score

For $z_i = \tilde{x}_i^\top \theta$, define $\text{softplus}(z) = \log(1 + e^z)$. Then

$$\log \pi_i = z_i - \text{softplus}(z_i), \quad \log(1 - \pi_i) = -\text{softplus}(z_i).$$

$$\ell_i(\theta) = -y_i [z_i - \text{softplus}(z_i)] + (1 - y_i) \text{softplus}(z_i).$$

$$\boxed{\ell_i(\theta) = \text{softplus}(z_i) - y_i z_i}, \quad J(\theta) = \frac{1}{n} \sum_{i=1}^n \ell_i(\theta).$$

This is binary cross-entropy expressed through the logistic score.

Which probabilities enter the likelihood?

Illustrative predictions for three conditionally independent observations:

i	y_i	π_i	Observed-label probability
1	1	0.8	0.8
2	0	0.7	$1 - 0.7 = 0.3$
3	1	0.4	0.4

$$\mathcal{L} = 0.8 \times 0.3 \times 0.4 = 0.096, \quad J = -\frac{1}{3} \log(0.096) \approx 0.781.$$

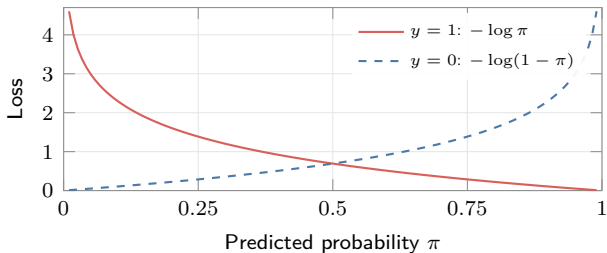
The second observation contributes $-\log(0.3)$, because its label is 0.

Response models and their fitting losses

	Gaussian linear regression	Logistic regression
Response family	Gaussian on $\mathbb{R}$	Bernoulli on $\{0, 1\}$
Conditional mean	$\tilde{x}_i^\top \theta$	$s(\tilde{x}_i^\top \theta)$
Conditional variance	σ^2	$\pi_i(1 - \pi_i)$
Fitting loss	Squared error	Binary cross-entropy

- A Bernoulli likelihood still needs a mean model. MLE does not force the logistic link.
- OLS is defined by minimizing squared error. Gaussian errors give it one likelihood interpretation.

A confident mistake incurs a large loss



$$\ell(y, \pi) = \begin{cases} -\log \pi, & y = 1, \\ -\log(1 - \pi), & y = 0. \end{cases}$$

For an observed positive label:

$$\pi = 0.9 \Rightarrow \ell \approx 0.105,$$

$$\pi = 0.01 \Rightarrow \ell \approx 4.605.$$

The loss uses probability confidence, not just the predicted class.

The derivative becomes a prediction residual

For one observation: $\ell = -y \log \pi - (1 - y) \log(1 - \pi)$ and $\pi = s(z)$

$$\frac{\partial \ell}{\partial \pi} = -\frac{y}{\pi} + \frac{1 - y}{1 - \pi}, \quad \frac{\partial \pi}{\partial z} = \pi(1 - \pi).$$

$$\frac{\partial \ell}{\partial z} = \left(-\frac{y}{\pi} + \frac{1 - y}{1 - \pi} \right) \pi(1 - \pi) = \boxed{\pi - y}.$$

The model probability minus the observed label drives the update.

From one observation to the whole sample

$$\nabla_{\theta} \ell_i = (\pi_i - y_i) \tilde{x}_i, \quad J(\theta) = \frac{1}{n} \sum_{i=1}^n \ell_i.$$

$$\boxed{\nabla J(\theta) = \frac{1}{n} X^{\top} (\pi - y)}, \quad \pi = s(X\theta).$$

	Prediction	Gradient of mean loss
Least squares	$X\theta$	$X^{\top}(X\theta - y)/n$
Logistic regression	$s(X\theta)$	$X^{\top}(\pi - y)/n$

The logistic objective is convex

$$\nabla^2 J(\theta) = \frac{1}{n} X^\top D(\theta) X, \quad D(\theta) = \text{diag}(\pi_i(1 - \pi_i)).$$

$$v^\top \nabla^2 J(\theta) v = \frac{1}{n} \sum_i \pi_i(1 - \pi_i) (\tilde{x}_i^\top v)^2 \geq 0.$$

- Every finite stationary point is a global minimum.
- Full column rank gives positive curvature at finite θ .

Convexity does not by itself guarantee that a finite minimizer exists.

Gradient descent for logistic regression

No general OLS-style closed form. Initialize $\theta^{(0)} = 0$; choose $\eta > 0$

$$\begin{aligned}z^{(t)} &= X\theta^{(t)}, \\ \pi^{(t)} &= s\left(z^{(t)}\right), \\ g^{(t)} &= \frac{1}{n}X^{\top}(\pi^{(t)} - y), \\ \theta^{(t+1)} &= \theta^{(t)} - \eta g^{(t)}.\end{aligned}$$

Monitor the objective and $\|g^{(t)}\|_2$; use a maximum iteration count.

One gradient step by hand

A separate three-observation example: $\theta^{(0)} = (b, w) = (0, 0)$ and $\eta = 1$

x_i	y_i	$\pi_i^{(0)}$
-1	0	1/2
0	0	1/2
1	1	1/2

Compute the average residual and its feature-weighted average.

$$g^{(0)} = \begin{bmatrix} 1/6 \\ -1/3 \end{bmatrix}, \quad \theta^{(1)} = \begin{bmatrix} -1/6 \\ 1/3 \end{bmatrix}.$$

$$J(\theta^{(0)}) = 0.69315 \longrightarrow 0.56688.$$

How large can the learning rate be?

Since $0 < \pi_i(1 - \pi_i) \leq 1/4$, the curvature has a global bound

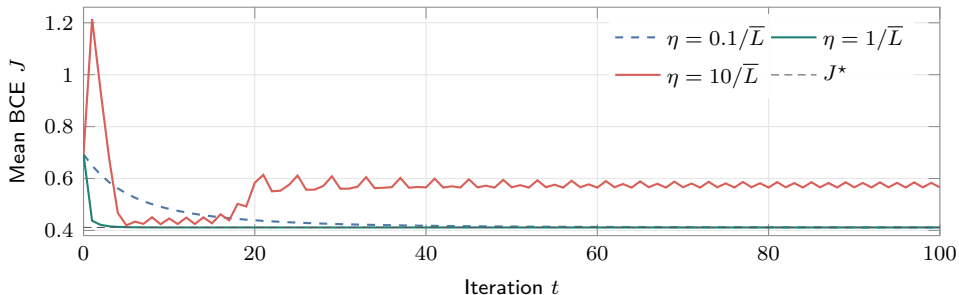
$$0 \preceq \nabla^2 J(\theta) \preceq \frac{X^\top X}{4n}, \quad \bar{L} = \frac{\|X\|_2^2}{4n}.$$

$$J(\theta - \eta g) \leq J(\theta) - \eta \left(1 - \frac{\bar{L}\eta}{2}\right) \|g\|_2^2.$$

$0 < \eta \leq 1/\bar{L}$ is a conservative choice for monotone descent.

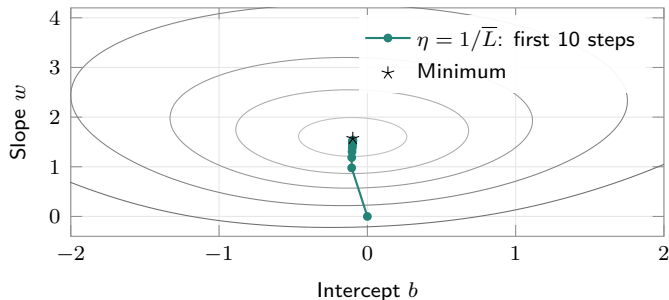
Loss curves: the step size changes the trajectory

The 20-point defect sample; feature $x = (x_{\text{raw}} - 5)/2$; $\theta^{(0)} = 0$



$\bar{L} = 0.47308$. All curves minimize the same mean binary cross-entropy.

Gradient descent moves across loss contours



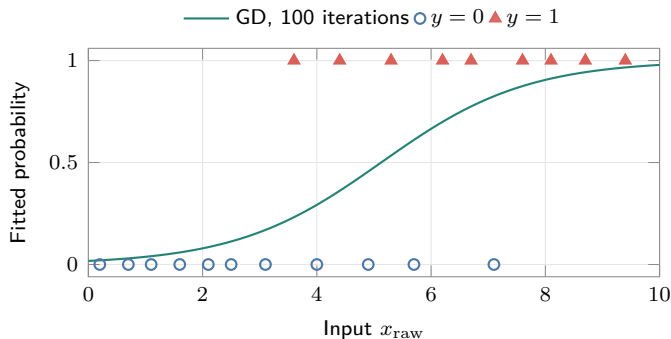
Each contour has the same mean loss.

Start: $(b, w) = (0, 0)$.

Shown: first ten updates with $\eta = 1/\bar{L}$.

The same run produces the green loss curve.

The optimized score gives a probability curve



Scaled input:

$$x = \frac{x_{\text{raw}} - 5}{2}.$$

Fitted score:

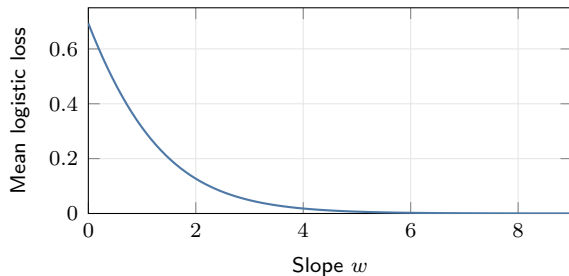
$$z = -0.09815 + 1.57004x.$$

$$J \approx 0.41137.$$

The labels overlap; the optimum has finite coefficients.

A good probability fit need not assign probability one to every observed label.

Separable data can send the MLE to infinity



Two observations: $(-1, 0)$ and $(1, 1)$; let $b = 0$.

$$J(w) = \log(1 + e^{-w}).$$

$$J(w) \downarrow 0 \quad \text{as } w \rightarrow \infty.$$

No finite w attains zero loss.

A vanishing loss can accompany exploding coefficients. Regularization can prevent this behavior.

Evaluate the loss without overflow

$$\ell(y, z) = \log(1 + e^z) - yz = \boxed{\max(z, 0) - yz + \log(1 + e^{-|z|})}.$$

- Compute cross-entropy directly from scores (logits).
- Avoid taking log of probabilities rounded to 0 or 1.
- Check training loss, held-out performance, and coefficient size.

The same likelihood can have stable and unstable numerical implementations.

One item, one defect label

An inspection system records the primary defect on each rejected item.

Class	Label	Possible visual evidence
1	Scratch	Elongated surface mark
2	Dent	Local indentation
3	Stain	Discolored region

The model assigns $\pi_k(x) = \Pr(y = k \mid x)$ for $k = 1, 2, 3$, with $\pi_1(x) + \pi_2(x) + \pi_3(x) = 1$.

One linear score for each class

For $x \in \mathbb{R}^{p-1}$, include the intercept in $\tilde{x} = (1, x^\top)^\top \in \mathbb{R}^p$.

$$z_k(x) = \tilde{x}^\top w_k, \quad W = [w_1 \ \cdots \ w_K] \in \mathbb{R}^{p \times K}.$$

The real-valued scores z_k are called **logits**.

$$\underbrace{Z}_{n \times K} = \underbrace{X}_{n \times p} \underbrace{W}_{p \times K}.$$

- Row i of X is $\tilde{x}_i^\top$.
- Each column of W describes one class.

Softmax probabilities

$$\pi_k(x) = \frac{\exp(z_k(x))}{\sum_{j=1}^K \exp(z_j(x))}, \quad k = 1, \dots, K.$$

- Finite logits give positive probabilities that sum to one.
- Every class shares the same denominator.

$$\hat{y}(x) = \arg \max_k \pi_k(x) = \arg \max_k z_k(x).$$

Predicting a label requires only the largest score. Likelihood training uses the entire normalized distribution.

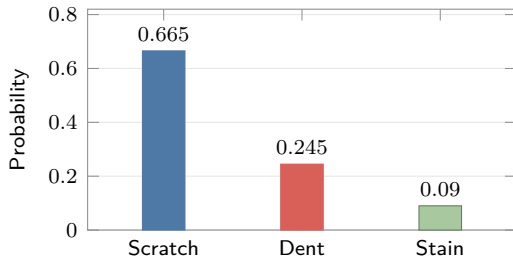
Three scores, three probabilities

Constructed example: $z = (2, 1, 0)^\top$.

Class	z_k	e^{z_k}	π_k
Scratch	2	7.389	0.665
Dent	1	2.718	0.245
Stain	0	1.000	0.090

The normalizer is 11.107.

A one-unit score advantage multiplies the probability ratio by e .



Categorical likelihood

Write $Y_{ik} = \mathbf{1}\{y_i = k\}$ and $P_{ik} = \pi_k(x_i)$.

$$L(W) = \prod_{i=1}^n \prod_{k=1}^K P_{ik}^{Y_{ik}} = \prod_{i=1}^n P_{i,y_i}.$$

$$\widehat{W} \in \arg \max_W \log L(W) = \arg \max_W \sum_{i=1}^n \sum_{k=1}^K Y_{ik} \log P_{ik}.$$

- The observed class selects one probability from each row.
- The product assumes conditionally independent labels given the inputs.

Cross-entropy and log-sum-exp

$$J(W) = -\frac{1}{n} \log L(W) = -\frac{1}{n} \sum_{i=1}^n \sum_{k=1}^K Y_{ik} \log P_{ik}.$$

For one observation with true class y ,

$$\ell(z, y) = \underbrace{\log \left(\sum_{k=1}^K e^{z_k} \right)}_{\text{LSE}(z)} - z_y.$$

For $z = (2, 1, 0)^\top$ and true class Dent, the loss is $-\log(0.2447) \approx 1.408$ nats.

The gradient is a probability residual

$$\frac{\partial \ell(z_i, y_i)}{\partial z_{ik}} = P_{ik} - Y_{ik}.$$

$$\underbrace{\nabla_W J}_{p \times K} = \frac{1}{n} \underbrace{X^\top}_{p \times n} \underbrace{(P - Y)}_{n \times K}.$$

- P applies softmax independently to each row of XW .
- Each class column accumulates feature-weighted probability errors.

Binary logistic regression is the $K = 2$ case

$$\pi_1(x) = \frac{e^{z_1}}{e^{z_1} + e^{z_2}} = \frac{1}{1 + e^{-(z_1 - z_2)}} = s\left(\tilde{x}^\top (w_1 - w_2)\right).$$

For any pair of classes,

$$\log \frac{\pi_j(x)}{\pi_k(x)} = z_j(x) - z_k(x) = \tilde{x}^\top (w_j - w_k).$$

Softmax regression makes every pairwise log odds linear in the features.

A reference class removes redundant parameters

$$\text{softmax}(z + c\mathbf{1}) = \text{softmax}(z), \quad c \in \mathbb{R}.$$

Adding the same $a \in \mathbb{R}^p$ to every column of W changes no prediction.

Choose class K as the baseline:

$$w'_k = w_k - w_K \quad (k < K), \quad w'_K = 0.$$

This leaves $(K - 1)p$ free coefficients.

- The fitted probabilities survive the change of coordinates.
- The baseline fixes this ambiguity, not rank deficiency or separation.

Stable computation uses score differences

For one row, set $m = \max_k z_k$.

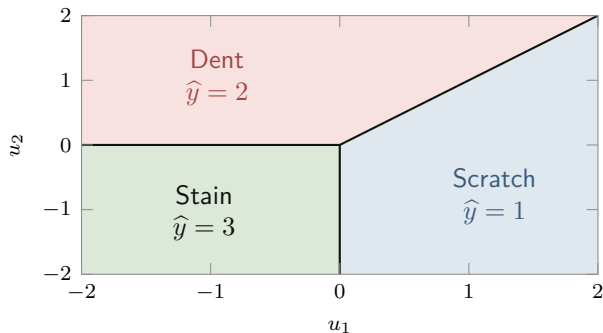
$$\pi_k = \frac{e^{z_k - m}}{\sum_j e^{z_j - m}}.$$

$$\ell(z, y) = (m - z_y) + \log \sum_j e^{z_j - m}.$$

Every exponential is at most 1, and at least one equals 1.

(1002, 1001, 1000) gives exactly the same probabilities as (2, 1, 0). Compute the loss from logits to avoid taking $\log(0)$ after underflow.

Three classes partition the feature plane



Constructed scores:

$$z_1 = u_1, \quad z_2 = u_2, \quad z_3 = 0.$$

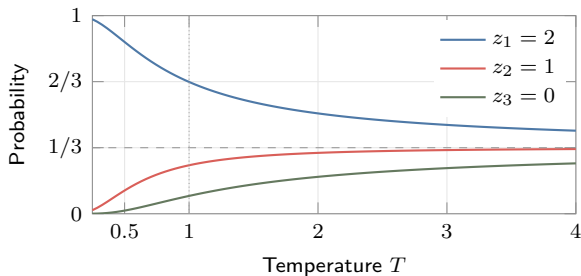
A boundary between winning classes j and k satisfies

$$\tilde{x}^\top (w_j - w_k) = 0.$$

Here $\tilde{x} = (1, u_1, u_2)^\top$.

Black segments mark ties for the largest score.

Temperature changes concentration



For fixed logits and $T > 0$,

$$\pi_k(T) = \frac{e^{z_k/T}}{\sum_j e^{z_j/T}}.$$

- Larger T spreads probability more evenly.
- The largest score and predicted class stay unchanged.

Softer probabilities do not automatically imply better calibration. Fit a calibration temperature on held-out labeled data.

Categorical outputs in vision and language

Image classification

$$\Pr(y = k \mid \text{image}) = \text{softmax}\left(A^\top h(\text{image}) + b\right)_k.$$

AlexNet used a 1000-class softmax output above learned visual features.

Next-token prediction

$$\Pr(v_t = k \mid v_{<t}) = \text{softmax}(z_t)_k, \quad J_{\text{seq}} = - \sum_t \log \Pr(v_t \mid v_{<t}).$$

The observed next token selects one vocabulary probability at each position.

Softmax also supplies attention weights

$$A = \text{softmax}_{\text{row}} \left(\frac{QK_{\text{key}}^{\top}}{\sqrt{d_k}} \right), \quad H = AV.$$

Use	Normalize over	Meaning of an entry
Output classifier	Classes or vocabulary	Probability of a target label
Attention row	Available key positions	Weight in a mixture of value vectors

Softmax attention is not itself a label-likelihood classifier. Its weights help construct the representations used for prediction.

One fitting principle, three response models

Choose a response model $\longrightarrow$ min average negative log-likelihood.

Model	Conditional response	Fitting objective
Linear regression	Gaussian, linear mean	Squared error
Logistic regression	Bernoulli, sigmoid mean	Binary cross-entropy
Softmax regression	Categorical, normalized logits	Multiclass cross-entropy

For logistic and softmax regression, the gradient compares predicted probabilities with observed labels.

References and further reading

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